University of Trento
Master's Degree in Computer Science

Data Visualization Proposal

Data Visualization Lab
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1 Context

1.1 Curiosity Origin

As frequent travelers, we are curious about urban mobility and how people navigate large cities. We became particularly interested in understanding mobility patterns in the context of sustainable transportation and green alternatives—recognizing their growing importance in urban planning and environmental goals. Specifically, we wanted to explore the underlying choices people make when selecting transportation modes and the factors that shape those decisions.

This curiosity led us to explore data sources that could provide meaningful insights into urban movement. Among them, New York City's bike rental data stood out as a rich source of information on how people travel across the city using a sustainable mode of transportation. When combined with other datasets, it can offer a clearer picture of the city's daily dynamics.

We also learned that this bike rental data is provided by a private company (`Citi Bike`) that operates the bike-sharing service in New York City. For this reason, making our analysis useful and relevant to the company is an effective way to increase its practical impact.

1.2 Project Circumstance and Purpose

`Citi Bike` currently collects vast amounts of operational data. If effectively analyzed and visualized, this data could provide valuable insights to optimize their services and enhance user experience.

Our visualization tool directly addresses these gaps by making complex mobility data accessible and actionable.

The primary objective is to transform raw `Citi Bike` operational data into clear, actionable insights for stakeholders.

1.2.1 Citi Bike Operations Team

Specifically, we aim to reveal:

1. **Temporal patterns**—peak usage hours, seasonal trends, and demand cyclicity.
2. **Spatial hotspots**—high-demand stations, commute corridors, and underutilized infrastructure
3. **User segmentation**—behavioral differences between members and casual riders

The first two insights can help the operations team optimize station placement and bike rebalancing. Meanwhile, the third can inform targeted marketing strategies and service improvements to better meet user needs.

1.2.2 City Planners and General Public

Beyond the operations team, our visualization can also benefit city planners by providing insights into commute flows and modal patterns, which can inform infrastructure investment and urban planning decisions. For the general public, it can raise awareness of urban mobility and sustainable transportation alternatives, fostering a more informed and engaged community.

1.3 Deliverables

1.3.1 Vision

We deliver an interactive web-based dashboard as the primary output. This allows all users to explore mobility patterns through a unified, user-friendly interface. Our design approach is to present clear, immediate insights at first impactful while also allowing deeper exploration for users who want to examine the details.

Regarding update frequency, we expect the dashboard to be refreshed automatically on a regular basis, since the data are continuously collected. This ensures that the insights remain useful and relevant over time, while also enabling historical analysis and comparisons across different periods.

1.3.2 Design Philosophy

Our visualization prioritizes clarity and analytical depth over visual polish. We employ a dual-level approach: high-level dashboards present immediate summary insights (e.g., daily ridership totals, top stations), while detailed views enable granular exploration (e.g., hourly breakdowns, user type comparisons). All design choices remain data-centric, avoiding decorative elements or misleading visual encodings that could obscure underlying patterns.

1.3.3 Resources

Since we are working with a large dataset and aim to create an interactive dashboard, we need programming skills and experience with data and visualization tools.

Our tech stack combines:

- **Backend (Python)** – efficient data processing, cleaning, and REST

API development via FastAPI;

- **Frontend (React/D3.js)** – custom, interactive visualizations tailored to mobility pattern analysis.

This modular architecture enables rapid iteration during development while remaining scalable as data volumes grow.

1.3.4 Constraints

In general, trying to obtain the most value from the data while respecting project constraints is a key challenge. The project must balance the ambition of performing meaningful analyses with practical limitations such as time availability, data quality, and implementation complexity. Since the final deliverable is scheduled for June, it is necessary to keep the feature scope focused and prioritize analyses that can provide the most relevant and actionable insights.

Another important aspect concerns the quality and consistency of the available data. Historical `Citi Bike` records were collected over multiple years, and they may contain missing values, inconsistencies, or differences in formatting and schema across datasets. For this reason, a careful preprocessing phase is required to clean, standardize, and integrate the data before any reliable analysis can be performed.

Finally, the large size of the dataset introduces additional technical challenges. Efficient aggregation, filtering, and data handling strategies must be implemented to manage the volume of information effectively. These optimizations are particularly important to ensure that the dashboard remains responsive and interactive, allowing users to explore the data smoothly without performance issues.

2 Proposed Datasets

Before selecting the datasets used in this project, we explored several mobility datasets from different cities in order to identify a suitable source of data. However, we found that many of these datasets were either incomplete, not regularly updated, or lacked the necessary granularity for our analysis.

After careful consideration, we decided to focus on the Citi Bike System Data for New York City, which is well maintained and reflects the complexity of the city.

2.1 Citi Bike System Data

The Citi Bike System Data is published by *Lyft*, a company that operates the bike-sharing service in New York City. Trip records, dating back to 2013,

are released monthly as open data, and they are available for download in CSV format.

Each trip record includes key attributes such as start and end times, start and end stations, and user type. These variables allow us to analyze temporal and spatial patterns in bike rentals.

Through API access, we can also obtain real-time data on bike availability and station status, which can be integrated into our dashboard for up-to-date insights.

2.2 Weather Data

Weather conditions can significantly influence urban mobility, so we plan to take into consideration the weather data.

To do this, we plan to use the Open-Meteo API, which provides historical weather data for any location worldwide, including New York City.

The dataset covers more than 80 years of observations with hourly temporal resolution and a spatial resolution between 1 and 11 km. As an open-source service, Open-Meteo offers a broad set of weather variables, allowing us to also evaluate how environmental conditions affect bike usage patterns.

2.3 New York City Datasets

During our exploration, we also considered various datasets available on the NYC Open Data platform. It offers a wide range of datasets, dashboards, and maps related to the city's transportation system, including bike lanes, traffic data, and other contextual information.

However, we want to keep our analysis focused only on the bike-sharing system, and the Citi Bike System Data already provides a comprehensive view of the city's bike rental patterns. Therefore, we decided to avoid adding unnecessary complexity by incorporating additional datasets from NYC Open Data, unless they are needed to provide further context or support specific insights.

3 Editorial Choices

Here we describe how these translate into concrete editorial choices the viewpoints we adopt, what we include or exclude, and how we guide the viewer's attention.

3.1 Viewpoints

We decompose the subject along four complementary dimensions, each motivated by its relevance to the target audience: As said before, we focus on the following dimensions:

- **Temporal dimension**
- **Spatial dimension**
- **User type** (member vs. casual)
- **Rideable type** (classic vs. electric)

Framing We restrict our analysis to completed trips with valid station coordinates and exclude rides longer than four hours, which our exploratory analysis identified as outliers likely caused by unreturned bikes. Weather data is integrated at the hourly level to contextualize usage fluctuations, but weather itself is not a primary viewpoint—it serves as an explanatory overlay rather than a standalone dimension. Likewise, we do not incorporate additional NYC Open Data sources unless a specific insight requires external context, in order to keep the narrative focused on the bike-sharing system.

Focusing We establish a visual hierarchy that mirrors the stakeholders' priorities.

Temporal and spatial patterns receive the most prominent placement in the dashboard, as they directly inform operational decisions like station placement and fleet management. Meanwhile, user type and rideable type are presented in secondary charts that support deeper dives into customer segmentation and product performance, but do not overshadow the core spatial-temporal analysis. Those are more relevant for marketing and customer experience teams, who can use those insights to tailor promotions and service offerings. However, at the same time, they result less actionable for operations teams, who are more interested in the overall demand patterns than the specific breakdowns by user or rideable type.

Meanwhile, weather data is integrated as a contextual layer within the temporal charts, allowing users to correlate demand fluctuations with weather conditions without dedicating separate space to weather analysis, which is not a primary focus of our project.

3.2 Early Design Considerations

While specific chart types and encodings will be refined during development, a few design decisions follow directly from the editorial choices above:

- **Map-centric spatial view**: Spatial data is inherently geographic, so an interactive map is the natural primary choice for our project. An heatmap layer can emphasize demand intensity at a glance.
- **Faceted temporal charts**: Temporal data we be used mainly to compare behaviour across different time periods, so faceted line charts or small multiples will allow us to show multiple time series (e.g., by user type) without cluttering a single chart.
- **Interactive filtering over static views**: Rather than producing separate dashboards for each dimension, we favour cross-filtering: selecting a time range, station, or user type in one chart updates the others, supporting exploratory analysis without overwhelming the interface.